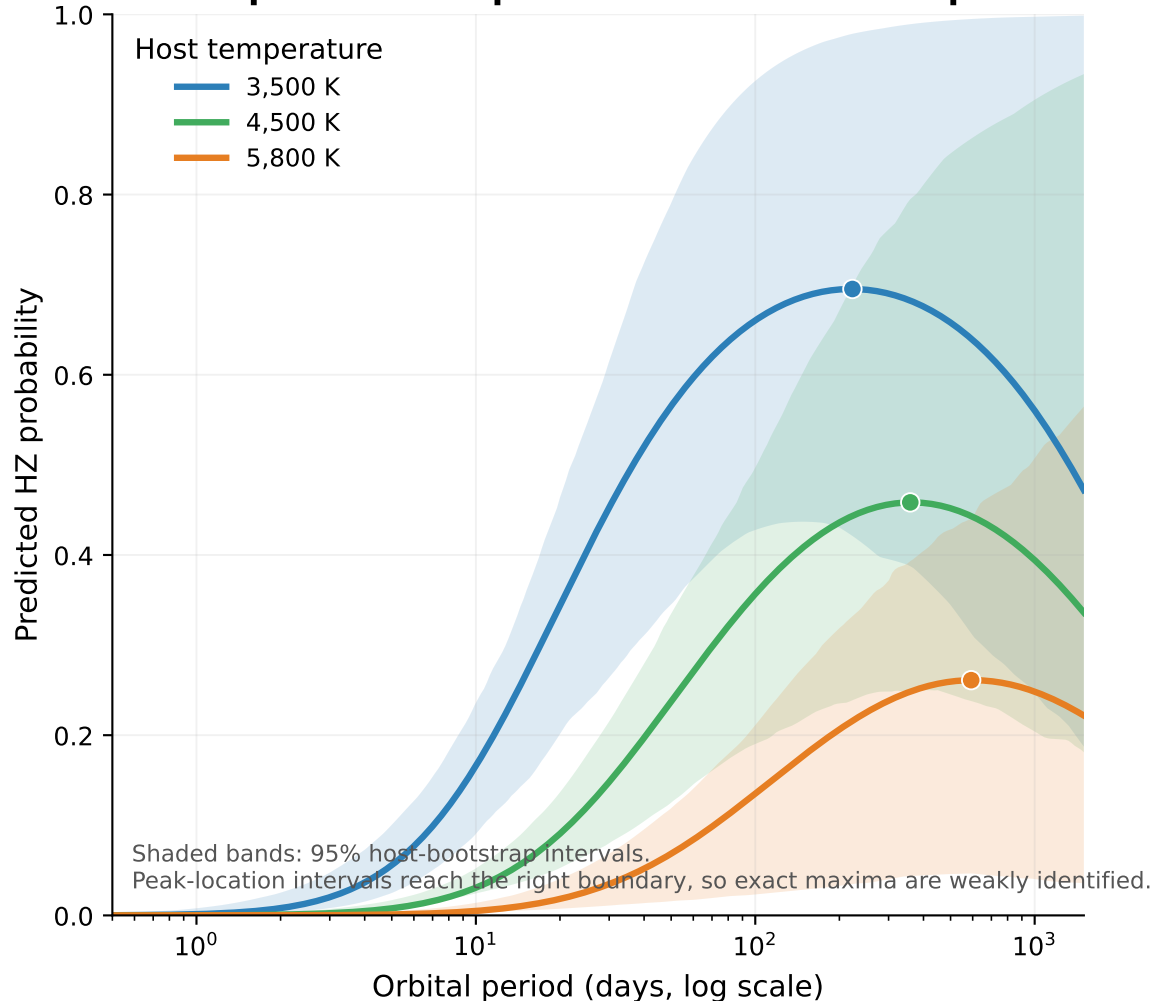


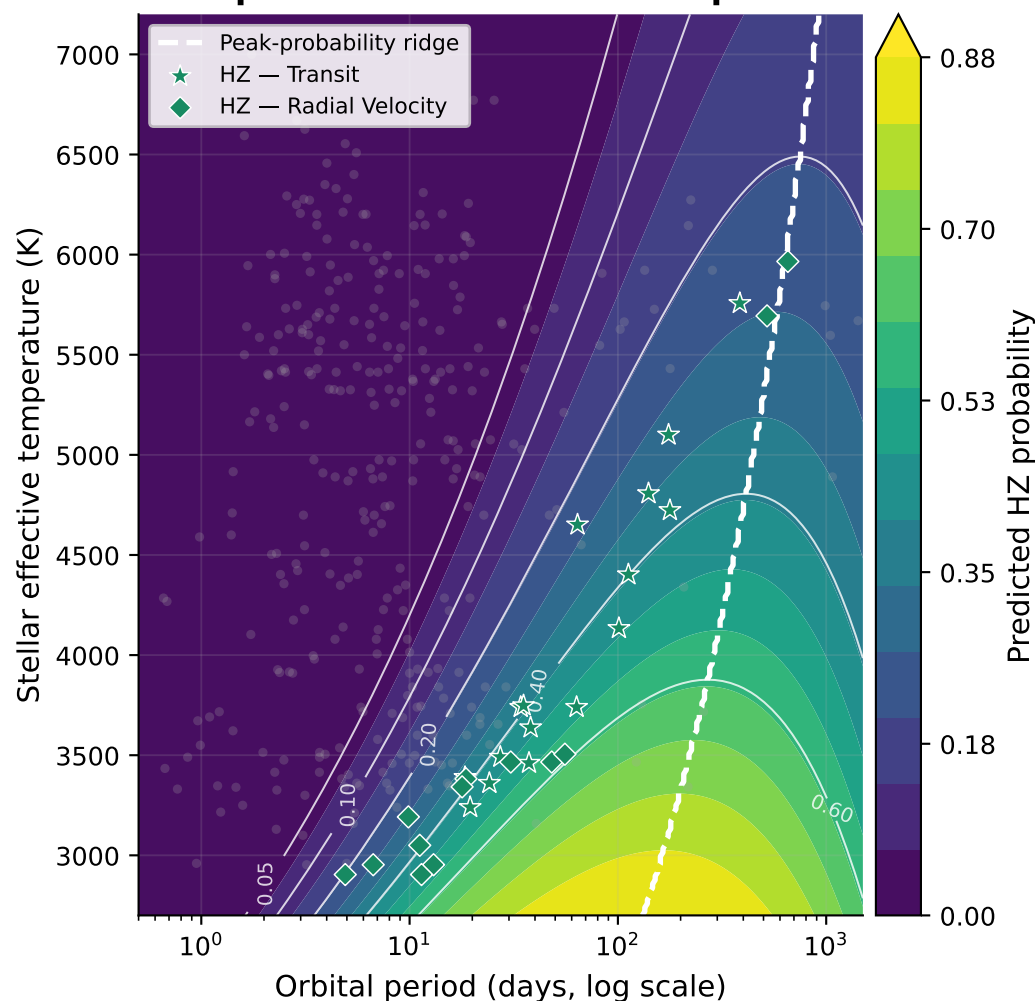
Model-implied period-temperature structure of HZ membership

Negative period curvature produces a moderate-period band; the positive interaction point estimate shifts that band toward longer periods for hotter stars, but the interaction coefficient remains uncertain.

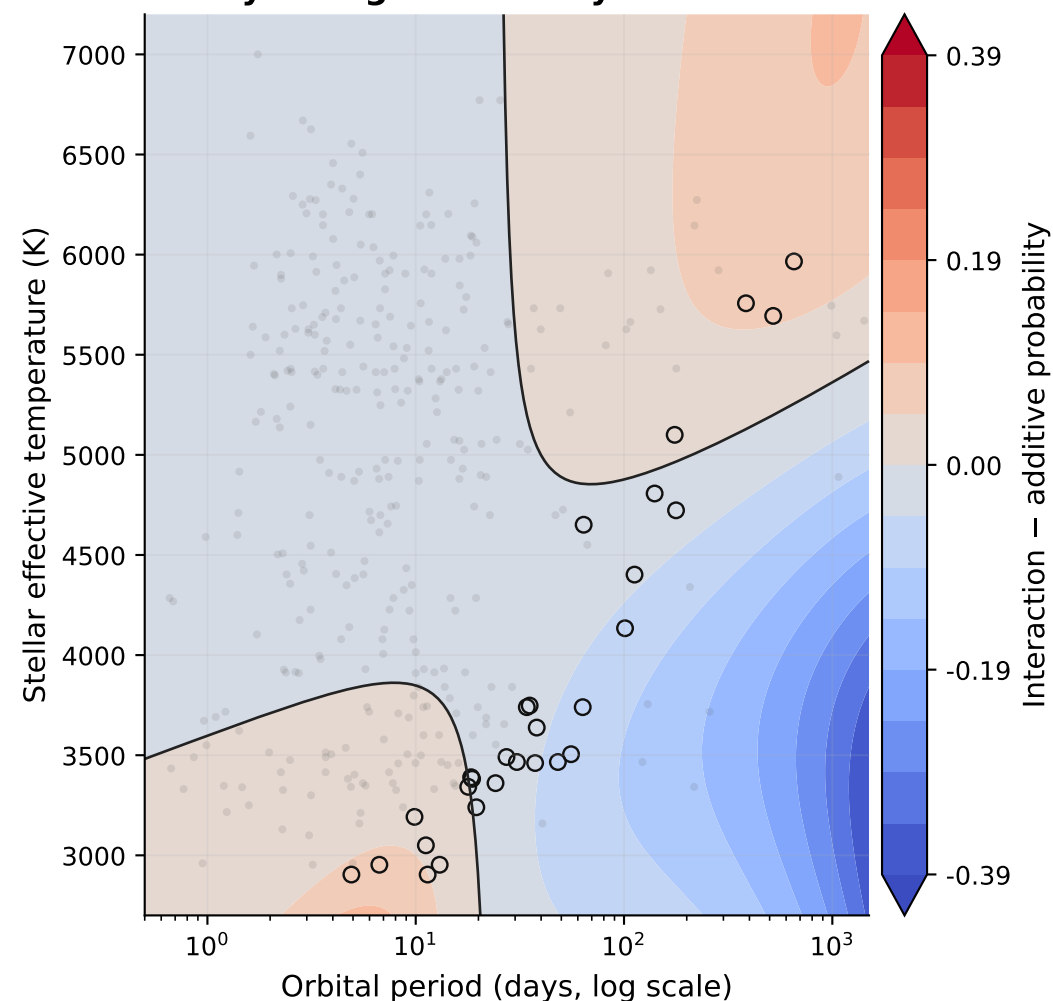
A. Period profiles at representative stellar temperatures



B. Full-sample interaction-model response surface



C. Probability change caused by the interaction term



These response surfaces are descriptive full-cohort fits; predictive performance is reported only from out-of-host cross-validation in the companion figure. This targeted model omits stellar radius, stellar mass, and discovery method, so exact peak periods must not be interpreted as physical HZ boundaries.